

## Practice

| Q.No. | Questions                                                                                                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | A junior software engineer signs the company's professional code of ethics but later ignores it when a client asks for a quick, insecure workaround to meet a deadline. Explain what "civic virtue" means in this context and why upholding it matters even under time pressure.                                                                           |
| 2     | Two CSE graduates argue about whether ethics is only about "not breaking the law." One says an act can be legal but still unethical. Give one real-world engineering example that supports this claim.                                                                                                                                                     |
| 3     | A team lead consistently arrives late to client meetings, causing the client to lose confidence in the team. Explain how this reflects poorly on the engineer's professional code of ethics, even though lateness is not illegal.                                                                                                                          |
| 4     | List three core skills needed for sound ethical reasoning in engineering decision-making, and briefly explain why each is necessary.                                                                                                                                                                                                                       |
| 5     | An engineering firm's code of ethics states "hold paramount the safety of the public." A project manager wants to skip a safety review to save time. Identify the ethical principle being violated and explain the manager's responsibility.                                                                                                               |
| 6     | Differentiate between "ethics" and "law" using one example from the software industry where the two might conflict.                                                                                                                                                                                                                                        |
| 7     | A newly hired engineer spends significant work hours on personal social media, delaying a critical deliverable. Explain this lapse from the standpoint of "valuing time" as a professional virtue.                                                                                                                                                         |
| 8     | Explain why civic virtue is considered a professional responsibility for engineers, and not merely a personal trait.                                                                                                                                                                                                                                       |
| 9     | A US-based software company outsources testing to a team in a country where small "facilitation payments" to speed up customs clearance of hardware are culturally normal, but such payments are illegal under the company's home-country law. Explain the ethical dilemma and suggest what the company should do.                                         |
| 10    | Explain the difference between legal responsibility and ethical responsibility, using an example where an engineer follows the law but still acts unethically.                                                                                                                                                                                             |
| 11    | A bridge collapses years after construction because the engineer skipped a load-test that was recommended by the original design but not legally mandated. Discuss whether this is a case of negligence, and explain the engineer's professional accountability.                                                                                           |
| 12    | A multinational engineering team spans three countries with different holiday practices and communication norms. Describe two cultural challenges this raises and one method to manage them respectfully.                                                                                                                                                  |
| 13    | Discuss one real engineering failure (structural or software) caused by ethical negligence, and identify the ethical principle that was violated.                                                                                                                                                                                                          |
| 14    | A remote engineer is asked to certify safety compliance for equipment they never personally tested, relying only on a colleague's claim that it passed. Explain why this is a violation of professional accountability.                                                                                                                                    |
| 15    | Explain what is meant by "negligence" in an engineering context, and how it differs from an honest mistake.                                                                                                                                                                                                                                                |
| 16    | An engineering consultant is legally permitted to design under lower safety standards in a foreign country, but doing so puts local workers at greater risk than in the consultant's home country. Should the consultant follow the local minimum standard or the stricter home-country standard? Justify your answer using ethical (not legal) reasoning. |
| 17    | An IT student copies a diagram and its explanation from a tutorial website into a project report, without giving credit. The instructor notices this while grading. Explain two actions the student should have taken instead, to stay professional.                                                                                                       |
| 18    | In a project, a team member says all tests passed, but two tests actually failed. Identify two professional behaviors the team member should have shown instead.                                                                                                                                                                                           |
| 19    | An engineer discovers that a colleague's design choice will cut costs but will increase environmental pollution beyond legal limits within a few years. Explain the engineer's "responsibility to society and the environment" in this situation.                                                                                                          |
| 20    | Two engineers disagree over a design approach, and the disagreement becomes personal, harming team communication. Suggest two respectful communication practices they should use to resolve the conflict professionally.                                                                                                                                   |
| 21    | A project team includes both mechanical and electrical engineers; a scheduling conflict arises because "done" means something different to each discipline. Explain one ethical challenge in cross-disciplinary collaboration this raises, and how it can be resolved.                                                                                     |
| 22    | Explain the difference between an "ethical dilemma" and a simple bad decision, giving one example of each.                                                                                                                                                                                                                                                 |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23 | An engineer holds shares in a supplier company and recommends that supplier's product for a project without disclosing the financial connection. Identify the type of ethical issue this represents and explain why disclosure matters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 24 | A junior engineer notices that a senior engineer takes credit for the junior's original idea during a client presentation. Discuss this situation in terms of integrity and honesty in the workplace.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 25 | Explain why "teamwork ethics" matters even when individual performance is measured separately, using one workplace example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 26 | <p>A cloud company earns revenue from user data. Its revenue is modeled as:<br/> <math>R(x) = 80x</math><br/> The privacy-risk cost of collecting this data is modeled as:<br/> <math>C(x) = x^2</math><br/> where <math>x</math> is the amount of data collected, in terabytes (TB).<br/> Separately, the same company discovers a security bug in its systems. If the bug is not fixed, there is a 15% chance of an attack costing \$80,000. Fixing it now costs \$6,000.</p> <p>(i) Write the profit function <math>P(x)</math>.<br/> (ii) Find the value of <math>x</math> that gives the maximum profit (use differentiation).<br/> (iii) Find the maximum profit.<br/> (iv) In one line, explain why collecting too much data is risky.<br/> (v) Find the expected loss if the company waits instead of fixing the bug.<br/> (vi) Compare the expected loss with the fix cost.<br/> (vii) Should the company fix the bug now or wait?<br/> (viii) In one line, explain why fixing it now is the ethical choice.</p>                                                                           |
| 27 | A hiring AI ranks some equally qualified candidates lower because of which university they attended. Name two ACM/IEEE ethics principles this breaks, and explain each in one line.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 28 | A developer finds that user passwords are stored without encryption. No breach has happened yet, and management does not know. State the first task the developer should perform. Name one ACM/IEEE principle that supports your answer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 29 | A mobile app collects users' location data "to improve service" but actually sells it to third-party advertisers without clear disclosure. Identify which data-ethics principle is violated and what the company should have done instead.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 30 | A company retains customer data indefinitely "just in case it's useful later," even after the customer closes their account. Explain the concept of "data minimization" and why it matters in this case.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 31 | An engineer is asked to add a hidden feature that logs users' keystrokes without informing them, described internally as "for debugging only." Explain the privacy concern and what the engineer should do.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 32 | Explain the difference between "data protection" (technical/legal safeguards) and "data ethics" (moral responsibility), using one example.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 33 | <p>An engineering firm's revenue from a manufacturing process is modeled as:<br/> <math>R(x) = 90x</math><br/> The environmental clean-up cost of running the process is modeled as:<br/> <math>C(x) = 2x^2</math><br/> where <math>x</math> is the production volume, in thousand units.<br/> Separately, the firm learns that its wastewater treatment system has a fault. If it is not repaired, there is a 25% chance of a pollution violation costing \$150,000 in fines. Repairing it now costs \$10,000.</p> <p>(i) Write the profit function <math>P(x)</math>.<br/> (ii) Find the value of <math>x</math> that gives the maximum profit (use differentiation).<br/> (iii) Find the maximum profit.<br/> (iv) In one line, explain why maximizing production without limits is environmentally risky.<br/> (v) Find the expected loss if the firm waits instead of repairing the fault.<br/> (vi) Compare the expected loss with the repair cost.<br/> (vii) Should the firm repair the fault now or wait?<br/> (viii) In one line, explain why repairing it now is the ethical choice.</p> |
| 34 | <p>A recruitment-tech company earns revenue by licensing its hiring AI to clients. Its revenue is modeled as:<br/> <math>R(x) = 70x</math><br/> The cost of auditing the AI for bias, as more clients are added, is modeled as:<br/> <math>C(x) = x^2</math><br/> where <math>x</math> is the number of client companies (in tens) using the AI.<br/> Separately, the company discovers the AI unfairly ranks candidates from certain universities lower. If this bias is not corrected, there is a 30% chance of a discrimination lawsuit costing \$90,000. Correcting the bias now costs \$7,000.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p>(i) Write the profit function <math>P(x)</math>.</p> <p>(ii) Find the value of <math>x</math> that gives the maximum profit (use differentiation).</p> <p>(iii) Find the maximum profit.</p> <p>(iv) In one line, explain why scaling an unaudited AI system to more clients is risky.</p> <p>(v) Find the expected loss if the company waits instead of correcting the bias.</p> <p>(vi) Compare the expected loss with the correction cost.</p> <p>(vii) Should the company correct the bias now or wait?</p> <p>(viii) In one line, explain why correcting it now is the ethical choice.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 35 | <p>A company sells an employee-monitoring software product. Its revenue is modeled as:<br/> <math>R(x) = 60x</math><br/> The cost of maintaining employee-privacy safeguards as more monitoring features are added is modeled as:<br/> <math>C(x) = x^2</math><br/> where <math>x</math> is the number of monitoring features enabled (in units).<br/> Separately, the company finds that one feature secretly records employees' private messages without consent. If this is not disabled, there is a 40% chance of a privacy complaint costing \$60,000 in penalties. Disabling the feature now costs \$4,000.</p> <p>(i) Write the profit function <math>P(x)</math>.</p> <p>(ii) Find the value of <math>x</math> that gives the maximum profit (use differentiation).</p> <p>(iii) Find the maximum profit.</p> <p>(iv) In one line, explain why adding more monitoring features without limits raises privacy risk.</p> <p>(v) Find the expected loss if the company waits instead of disabling the feature.</p> <p>(vi) Compare the expected loss with the disabling cost.</p> <p>(vii) Should the company disable the feature now or wait?</p> <p>(viii) In one line, explain why disabling it now is the ethical choice.</p> |
| 36 | <p>A construction company's revenue from a building project is modeled as:<br/> <math>R(x) = 120x</math><br/> The cost of additional structural safety reinforcement is modeled as:<br/> <math>C(x) = 3x^2</math><br/> where <math>x</math> is the number of safety-reinforcement units added.<br/> Separately, an inspector finds a load-bearing defect in the design. If it is not corrected, there is a 15% chance of a structural failure costing \$500,000 in damages and liability. Correcting it now costs \$15,000.</p> <p>(i) Write the profit function <math>P(x)</math>.</p> <p>(ii) Find the value of <math>x</math> that gives the maximum profit (use differentiation).</p> <p>(iii) Find the maximum profit.</p> <p>(iv) In one line, explain why cutting back on safety reinforcement to raise short-term profit is risky.</p> <p>(v) Find the expected loss if the company waits instead of correcting the defect.</p> <p>(vi) Compare the expected loss with the correction cost.</p> <p>(vii) Should the company correct the defect now or wait?</p> <p>(viii) In one line, explain why correcting it now is the ethical choice.</p>                                                                                |
| 37 | <p>A social media company earns revenue by selling anonymized user data to advertisers. Its revenue is modeled as:<br/> <math>R(x) = 55x</math><br/> The legal-compliance cost of managing this data is modeled as:<br/> <math>C(x) = x^2</math><br/> where <math>x</math> is the amount of user data sold, in millions of records.<br/> Separately, the company discovers that some "anonymized" records can actually be traced back to individuals. If this flaw is not fixed, there is a 35% chance of a data-privacy lawsuit costing \$200,000. Fixing the anonymization flaw now costs \$12,000.</p> <p>(i) Write the profit function <math>P(x)</math>.</p> <p>(ii) Find the value of <math>x</math> that gives the maximum profit (use differentiation).</p> <p>(iii) Find the maximum profit.</p> <p>(iv) In one line, explain why selling more data without proper anonymization increases risk.</p> <p>(v) Find the expected loss if the company waits instead of fixing the flaw.</p> <p>(vi) Compare the expected loss with the fix cost.</p> <p>(vii) Should the company fix the flaw now or wait?</p> <p>(viii) In one line, explain why fixing it now is the ethical choice.</p>                                        |